Team Details

Team Vexora

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Problem statement Id : **ps 7**

Problem Statement description: **One Patient, Many Records**

Problem & Why Current Solutions Fall Short

Scattered history ,Separated platforms, dangerous AI summarizers

THE PROBLEM

1. **Fragmented Records**
   History is scattered across clinics, labs and hospitals.
2. **Repeats, Conflicts, Staleness**
   Duplicates and outdated entries pile up over time.
3. **Patient & Caregiver Burden**
   Families stitch records together by hand.

WHERE CURRENT TOOLS FAIL

1. **Single-Network Portals**
   Only show one provider’s records.
2. **Naive AI Summaries**
   Merge old and current medicines, drop dates, hide conflicts.
3. **False-Negative Hazard**
   Read “not documented” as “does not have.”

Our Solution

Stays on the device

Cloud: anonymized data only

1

Local On-Device Extraction

Documents are read and parsed on your device by our Python OCR

2

Automated PII Shield

Before anything leaves the device, sensitive PII is stripped and masked.

3

Groq Cloud Analysis

Anonymized text/JSON goes to Groq's LPU running gpt-oss-120b.

4

Doctor-Ready Summary

Groq returns an organized clinical summary and insights for physician review.

Contradiction Flagging

Evidence-Linked Contradiction Flagging

1. **Anonymized Timeline**
   Local Python engine compiles all records into one safe, structured JSON history.

2. **Strict AI Guardrails**
   Groq's rule: "Never guess, invent facts, or silently resolve conflicts."

3. **Smart Cross-Referencing**
   Catches mismatches, like a drug the patient is allergic to, or conflicting doses.

4. **Source-Linked Alerts**
   Warning flags for the doctor, linked to the exact source document IDs.

CONTRADICTION FLAG

Allergy on record

Penicillin allergy: source doc A

Conflicts with

Amoxicillin prescribed: source doc B

Doctor to review before dispensing.

Impact & Benefits

Transforming Care Across Patients, Doctors & Caretakers

Patients

Own their story

- No more paper chaos
- Transparent ownership of full health data
- Privacy-first architecture
- Instant guidance on what medicines are for and why they changed

Doctors

Clinical workspace

- Connected patient directory
- Targeted record uploads
- Immutable HIPAA-grade audit logs
- Speech-to-text voice assistant with real-time medication pop-ups.

Family Caretakers

Care without risk

- Monitor dependent family wards
- Track medication checklists
- Zero-risk, reversible delegation

Scalability & Future Scope

Scalability

Cost-efficient multi-tier scaling

1. Individuals
2. Clinics
3. Families

Future Scope

RAG for infinite history (pg vector)

Decades of records exceed token limits. A local vector database will retrieve only the chunks relevant to a doctor's query.

On-device WebGPU LLM

Local inference for fully air-gapped environments.

Global Medical Standards & Digital IDs

Sharing medical records securely across borders using global standards and unalterable digital IDs.